Tectonophysics, 63 (1980) 75—90

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FURTHER PALAEOMAGNETIC CONFIRMATION OF THE MAGELLANES OROCLINE

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ABSTRACT

Burns, K.L., Rickard, M.J., Belbin, L. and Chamalaun, F., 1980. Further palaeomagnetic confirmation of the Magellanes orocline. In: M.R. Banks and D.H. Green (Editors), Orthodoxy and Creativity at the Frontiers of Earth Sciences (Carey Symposium). Tectonophysics, 63: 97—90.

Palaeomagnetic determinations for Jurassic volcanics and a Cretaceous quartz—dolerite sill at three localities around the arcuate Patagonian Cordillera show that a bending of 40° to 27° has occurred. Since the structural bend is 90° it is probable that the Mesozoic arc was originally curved. Cainozoic dykes and sills at Lago San Martin in the northward-trending belt and Sierra Valdivieso in the eastward-trending belt have the same magnetic directions and were intruded after bending.

These results essentially confirm the palaeomagnetic determinations of Dalziel et al., (1973a) and further suggest that the collapse of the marginal sea during the Late Cretaceous orogenesis resulted in greater bending in the outer arc than in the inner arc. The orocline must have formed during the Andean orogeny between 80 m.y. and 21 m.y. ago and was accompanied by folding and thrusting.

New K—Ar dates for the Dientes de Navarino Sill (105 m.y.), basalt at Isla Carlos in the Straits of Magellan (21 m.y.) and diorite and andesite sills at Lago San Martin (14 m.y.) are also presented.

INTRODUCTION

Carey (1958) first proposed that the abrupt curving in the strike of the Cordillera from N—S in Patagonia to E—W in Tierra del Fuego may be due to actual bending. This paper reports the work of a small joint expedition from the Australian National University and Macquarie University, which, in the 1968—1969 summer season, set out to determine whether bending had occurred and the structural implications of such movements.

The general geology of this part of South America was established by

Feruglio (1949) and Krank (1932). Geologists of the Empresa National del Petroleo, particularly S. Cespedes, have mapped in some detail all the Cainozoic and Cretaceous rocks in the foreland zone of the inner side of the curving Cordilleran arc and have made reconnaissance traverses in the deformed and metamorphosed core zone along the coasts of the fiords in nothern and western Tierra del Fuego and Magellanes. This work is reported mainly in unpublished company reports but some of the details and general summaries are published by Gonzalez (1965) and Katz (1964a, b, c, 1972).

More recently the Lamont-Doherty Geological Observatory has been undertaking a continuing research project in the whole Scotia Arc region. Their work suggests that the palaeogeographic situation in the Cretaceous was that of a marginal basin (Dalziel et al., 1973b) with a seaward (south-westward) volcanic arc along the present pluton-studded metamorphic core zone, and that the whole Scotia Arc has had a similar and related history (Dalziel and Cortes 1972, Dalziel and Elliot, 1973). Halpern (1973) has dated many of the important rock units in this area.

Palaeomagnetism provides the only method of proving bending. Early magnetic determinations in the Antarctic Peninsula (Blundell, 1962) were interpreted as indicating bending by Hamilton (1966). Dalziel et al. (1973a), on the basis of Blundell's and new data, concluded that the hypothesis was unjustified and that the shape of the Antarctandean Cordillera between 68° S and 68° 30'S has not changed significantly since the Late Cretaceous. Dalziel et al., (1973a) also sampled Andean granodiorite plutons in the outer Magellanes Arc and showed that the magnetic directions are deflected around the bend, whereas those for Cainozoic dykes are not. Our work concentrated on sampling the Jurassic volcanics of the "Serié Tobifera" around the inner side of the arc (Figs. 1 and 2), although samples of Cretaceous and Tertiary dykes and sills were also collected. As part of this work, the Dientes de Navarino sill, basalt flows at Isla Carlos in the Straits of Magellan, and diorite and andesite sills at Lago San Martin, were dated by K-Ar method (Table I).

Hand specimens collected for palaeomagnetic work were oriented by Brunton or by sun compass where possible; sampling sites are indicated on Fig. 1. Measurements on the Navarino and some of the Lago San Martin samples were carried out by Chamalaun using an astatic magnetometer; later Belbin measured the rest of the collection and remeasured the weaker samples on the more sensitive Digico balanced fluxgate rock magnetometer. Cleaning was mainly by AF demagnetization but sedimentary specimens were subjected to thermal cleaning. Stereograms of magnetic directions from each locality are plotted in Fig. 3 and means are listed in Table II, a summary map of the aximuths of these directions and the results obtained by Dalziel et al. (1973a) are presented in Fig. 4.

TABLE I

K—Ar age determinations [1]

Sample	Locality	Mineral dated	%K	$*Ar^{40}/K^{40}$	% Ar^{40}	atm. age ($\times 10^6$ m.y.)
5. Biotite-augite, olivine, diorite	Lago San Martin	Biotite	7.11 7.16	0.0008233	21.1	14.0 ± 0.5
16. Andesite	Lago San Martin	Hornblende	0.490 0.493	0.0008293	72.5	14.1 ± 0.5
47. Basalt	Isla Carlos	Whole rock	1.10 1.10	0.001250	11.5	21.3 ± 0.6
121. Dolerite	Isla Navarino	Pyroxene	0.0936 0.0972	0.006912	70.6	115 ± 5

[1] Determinations by Dr. A. Webb of the Australian Mineral Development Laboratories. * denotes radiogenic 40 Ar.
Constants used: K^{40} = 0.0119 atom %
= $4.72 \cdot 10^{-10}$ y^{-1}
= $0.584 \cdot 10^{-10}$ y^{-1}

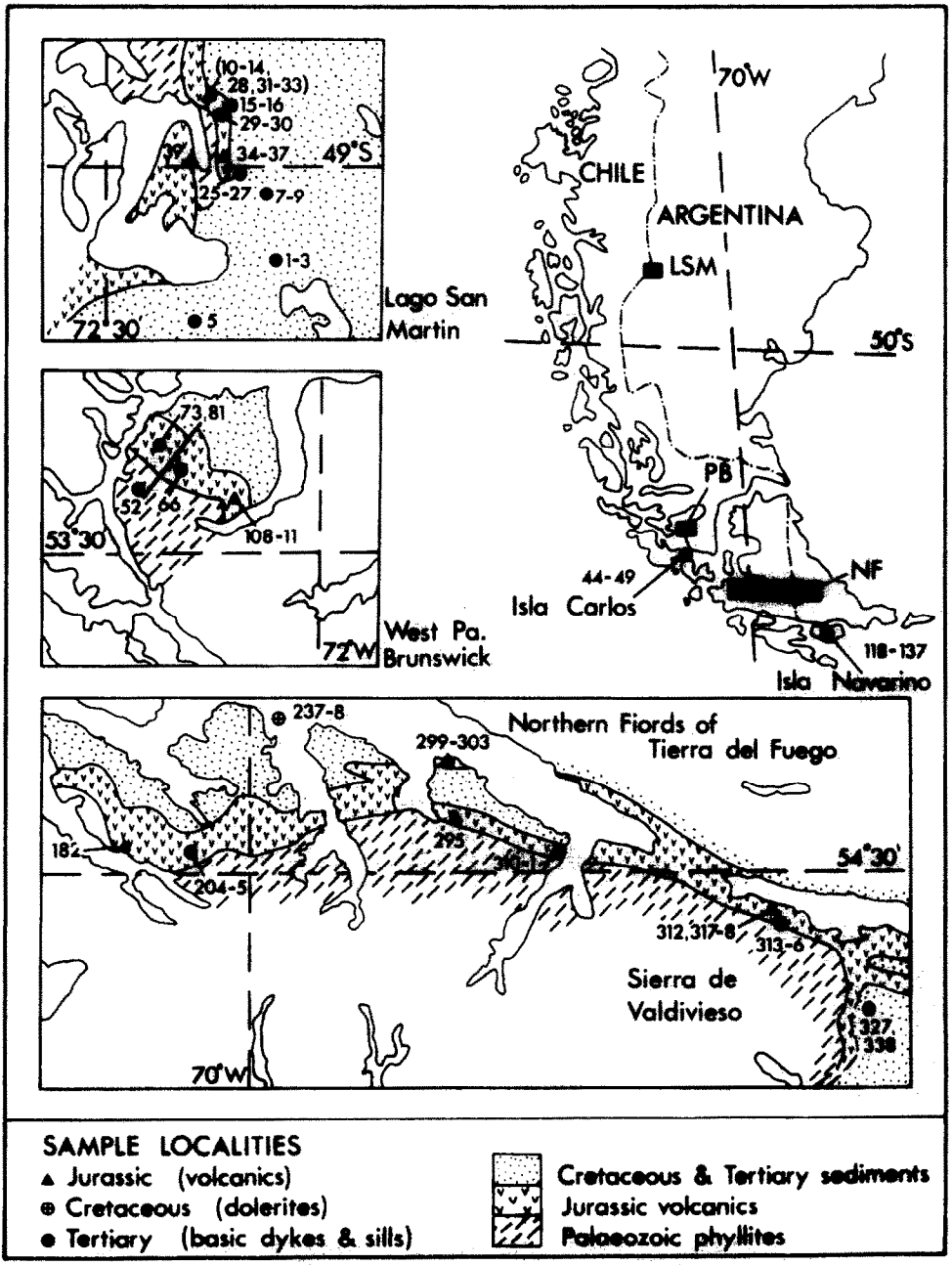


Fig. 1. General Geology of Patagonian Cordillera and sample localities.

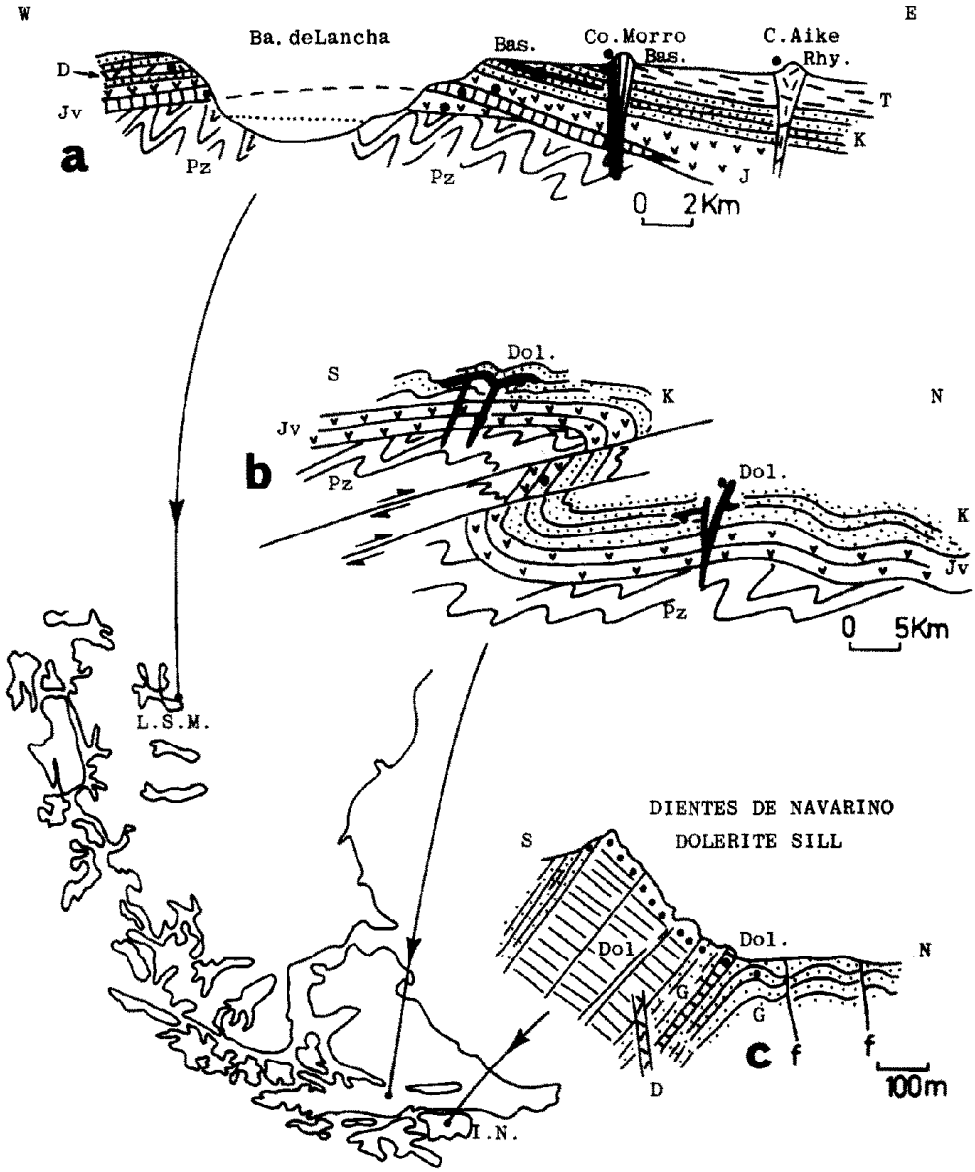


Fig. 2. Diagrammatic geological sections at sample localities. a. Bahia de Lancha-Lago San Martin, Agentina. Jurassic volcanics (*Jv*) overlying folded Palaeozoic greywackes (*Pz*) and shales. Cainozoic intrusions consist of diorite (*D*) and dolerite (*Dol.*) sills, and basalt (*Bas.*) and rhyolite (*Rhy.*) plugs. *J* = Jurassic; *K* = Cretaceous; *T* = Tertiary. b. Diagrammatic representation of the structure through the east-trending belt. c. Dientes de Navarino sill Navarino Island, Chile. Dolerite sill (*Dol.*) in Cretaceous Yahgan greywacke (*G*) cut by a small diorite (*D*) dyke. Position of sampling sites — ●; *f* = fault.

GENERAL GEOLOGY OF SAMPLING SITES

Mapping and sampling were carried out at four main localities around the arc (Fig. 1): the northward-trending belt at Bahia de Lancha, Lago San Martin, the bend itself at the western end of Peninsula Brunswick, and the eastward-trending belt along the fiords of northern Tierra del Fuego and at Navarino Island. The geology is depicted on a map in Fig. 1 and in sections in Fig. 2.

From Peninsula Brunswick southeastwards, the outcrop of Jurassic volcanics marks the hinge line of a major recumbent fold, broken by gently dipping thrusts (Fig. 2b), that swings with the strike around the bend. The sense of overturn is eastwards and northwards, towards the foreland. Cretaceous slates, shales and sandstones, followed by Cainozoic sediments, outcrop to the east and north on the inner side of the arc (Katz, 1964a, c). Axial planes of folds become upright over the foreland but they continue to reflect the swing of the arc. Cainozoic sills and plugs of dolerite, diorite, andesite and rhyolite intrude the older rocks especially at Lago San Martin (Fig. 2a), and in the Sierra de Valdivieso previously unreported sill swarms intrude the Cretaceous shales (Fig. 2b).

To the west and south of the arc, complexly folded Palaeozoic phyllites, forming the core of the orogen, lie unconformably beneath the Jurassic volcanics. These are intruded by Cretaceous granodiorites, and metamorphism has converted the phyllites to schists and gneisses in places. The Cretaceous "Yahgan" greywacke–shale (flysch) sequence is of different facies and crops out to the south of these metamorphics in Navarino Island and Isla Hoste where it is intruded by thick quartz–dolerite sills (Katz and Watters, 1966). The strikes are east–west here, but to the southwest the main pluton belt trends southeastward towards Cape Horn (Krank, 1932; Dalziel et al., 1973b).

PALAEOMAGNETIC DETERMINATIONS

Jurassic samples

The Jurassic volcanics at Lago San Martin consist largely of quartz–feldspar porphyries in welded ignimbrite sheets, a variety of acid tuffs, and, near the unconformable base of the sequence, some red tuffs and sandstones. To the south our collection consists mainly of pophyries with some flow-laminated rhyolites that are generally weakly magnetized. The porphyries commonly show columnar cooling joints and the attitudes of these and the flow laminations were used to make tectonic dip corrections.

At Lago San Martin three groups of gently dipping rocks were sampled; seven red tuffs and porphyries (Specs. 10–14, 28 and 32), from the northern area and three columnar porphyries (Specs. 36–39) and one red sandstone (Spec. 34) from the southern part of the area (Figs. 1 and 2a). Magnetic

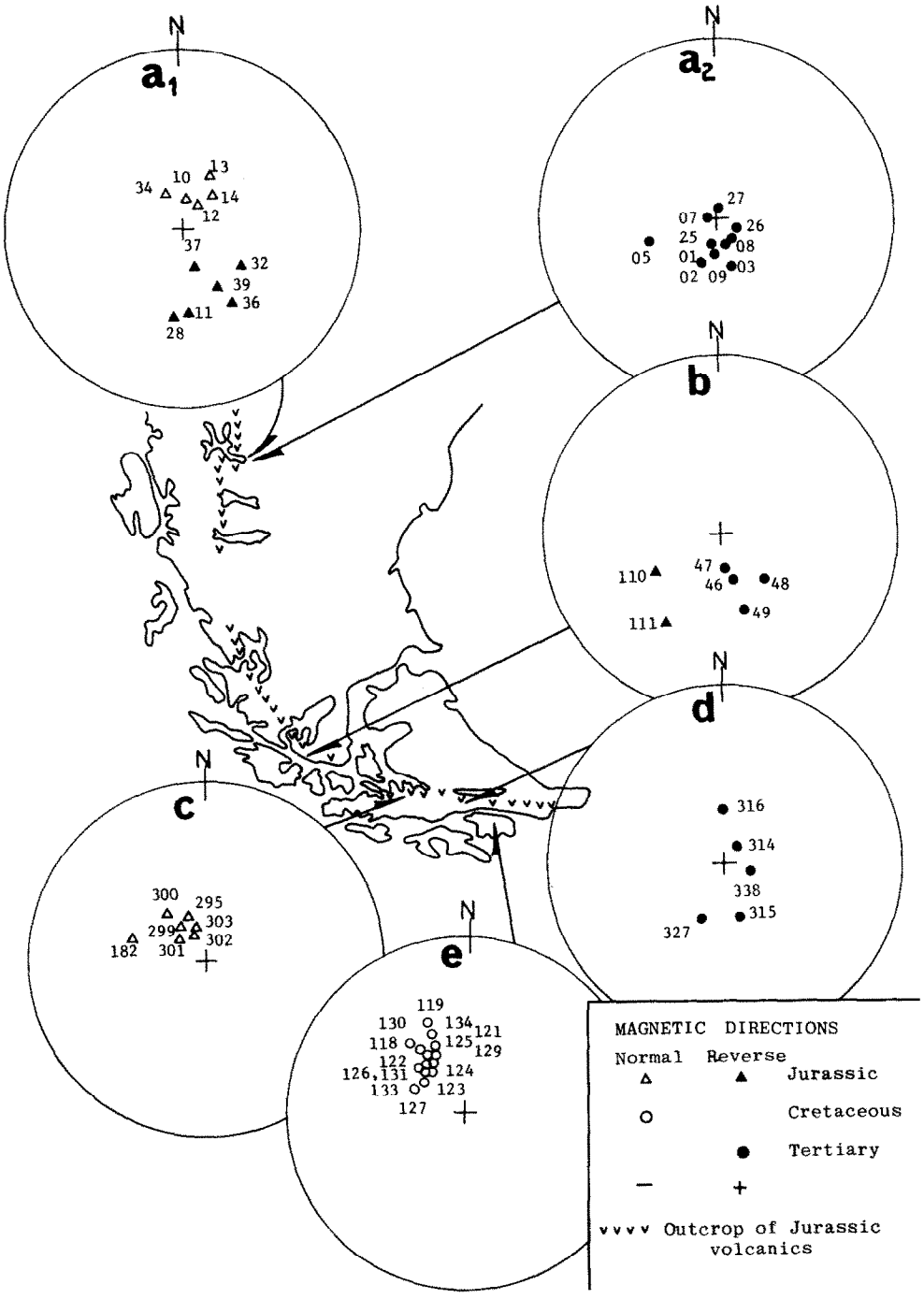


Fig. 3. Palaeomagnetic directions for individual samples. South-pole directions are plotted; reverse^{+} on lower hemisphere and normal^{-} on upper hemisphere. a_1. Lago San Martin — Jurassic. a_2. Lago San Martin — Cainozoic. b. Peninsula Brunswick — Jurassic, and Isla Carlos — Cainozoic. c. Northern Fiords of Tierra Del Fuego — Jurassic. D. Sierra Valdivieso — Cainozoic. e. Isla Navarino — Cretaceous.

TABLE II

Palaeomagnetic measurements

A.N.U. Geology Dept. Number (39001+) and locality	Rock type	Age	Key to Figs. 3, 4	No. of samples	No. of cylinders	Mean Dec.	Mean Inc.	α_{95}	(K)	(R)	VGP * (Fig. 4D)
Lago San Martin											
Cerro Morro											
7–9 (73°08′W, 49°02′S)	alkali basalt plug	Tertiary	a	3	6	169	78	18.02	47	2.958	
Cerro Cach Aike											
1–3 (73°05′W, 49°09′S)	rhyolite plug	Tertiary	a	3	6	182	62	14.779	70	2.972	
5 (73°15′W, 49°12′S)	diorite sill	14 m.y.	a	1	2	252	47	10.0	0.63	1.998	
25–27 (73°10′W, 49°02′S)	basalt sills	Tertiary	a	3	9	155	82	21.115	35	2.948	
	All intrusives	Tertiary	Mean a	10	23	189	74	11.4	18	9.523	(167, 267)
10–14 (72°13′W, 48°53′S)	volcanics	Jurassic	a	6	25	159	46	15.4	20	5.750	
28 (72°14′W, 48°53′S)											
31–33 (72°14′W, 48°52′S)											
36–39 (72°15′W, 49°01′S)											
34 (72°12′W, 48°59′S)			a	5	17	021	−68	12.63	37	4.894	
		Jurassic	Mean a	11	42	352	−57	12.90	13	10.260	(167, 283)
Isla Carlos											
46–49 (72°04′W, 53°45′S)	basalt	21.3 m.y.	b	4	9	160	55	16.31	32	3.908	
Northern Fiords											
182 (70°25′W, 54°27′S)											
295 (69°29′W, 54°24′S)	volcanics	Jurassic	c	7	23	316	−64	10.423	34	6.826	(151, 020)
299–303 (69°25′W, 54°22′S)											
Sierra de Valdivieso											
314–316 (68°45′W, 54°35′S)											
327 (68°40′W, 54°36′S)	dolerite	Cretaceous	d	5	13	155	83	31.3	6	4.422	(156, 306)
338 (68°30′W, 54°41′S)											
Dientes de Navarino											
118, 121–127, 129–131, 133–134 (67°38′W, 53°01′S)	dolerite	115 m.y.	e	13	29	319	−52	5.153	65	12.85	(144, 043)

* V.G.P. given as colatitude, east longitude.

directions from both northern and southern localities fall into two distinct groups (Fig. 3a$_1$) a normal group of five samples trend 023° at −68° and a reversed group of six samples have a mean, trending 159° at 46°. These are not antiparallel and a similar phenomenon has been reported by Creer et al. (1972) from the Middle Jurassic Chon Aike Formation from Camerones Bay to the north of this area, although the polarities are opposite to those at

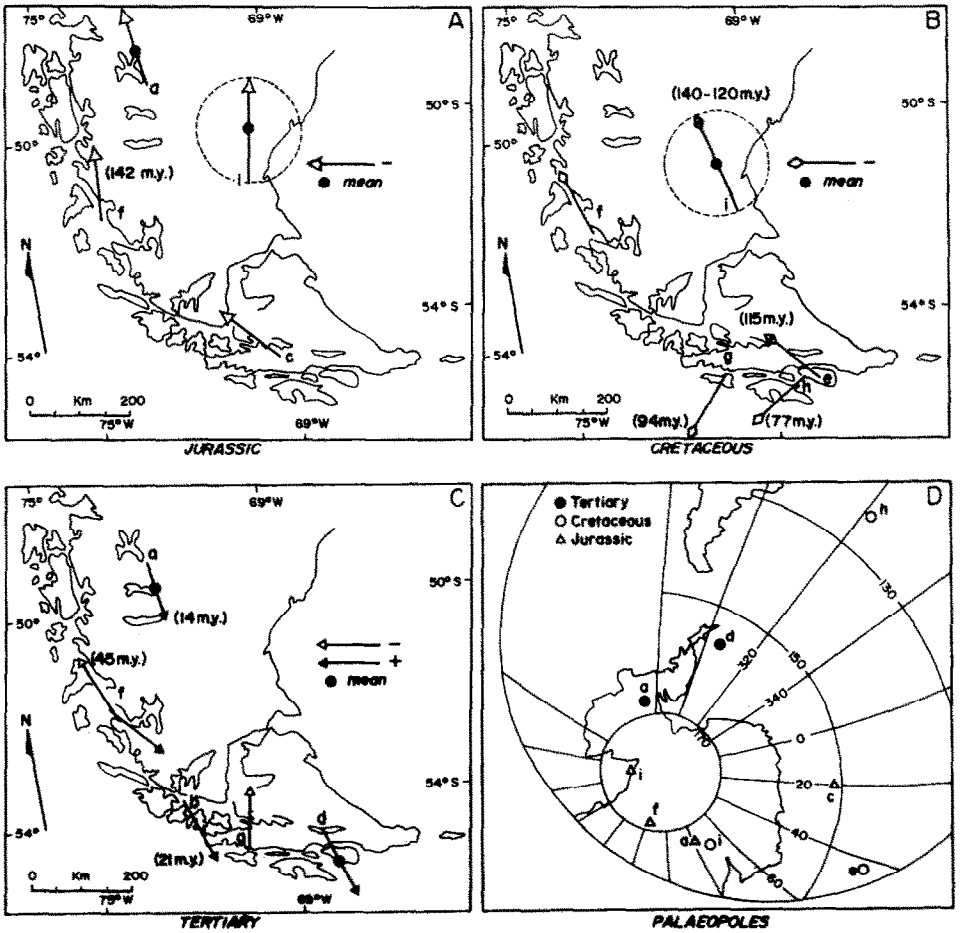


Fig. 4. Map of azimuths of magnetic-direction means around the Magellanes orocline. A. Jurassic; B. Cretaceous; C. Tertiary; D. Shows Jurassic and Cretaceous results plotted as virtual geomagnetic S-poles. Inner-arc results (*a*—*e*) this paper (Table II). Outer-arc results (*f*—*h*) from Dalziel et al. (1973a): *f* = Isla Rennel vicinity (Specs. SM33, SM41, SM53), Jurassic V.G.P. (172, 116); *g* = N.W. Isla Londonderry (Spec. WB2); *h* = Puerto Santa Rosa—N.W. Isla Navarino (Spec. BC89), Cretaceous V.G.P. (123, 321). *i* Inset: Jurassic (166 m.y.) magnetic direction mean reduced from Chon Aike Formation (Vilas, 1974), V.G.P. (175, 197); Cretaceous magnetic direction reduced from Serra Geral, Brazil (Creer, 1962, 1970), V.G.P. (168, 054).

Lago San Martin. The reversal is possibly associated with the Pospelovakaja Reversed Zone.

A site mean was calculated by inverting all the reversed polarities and adding them to the normal directions; this gives a magnetic direction of 352, —57 (Table II). Similar results are given by the two other methods for calculating a mean: (a) meaning the two cluster means: and (b) meaning the virtual geomagnetic poles obtained from each cluster and reducing to a magnetic direction.

A fold test only improved the Fisher dispersion parameter from $K = 8$ to $K = 13$ which is insignificant at the 90% level (McElhinny, 1964). This is probably because of the low dips and the possibility that there was an initial dip in the volcanic layers.

Vilas (1974) has calculated a mean pole at 85S, 197E ($\alpha_{95} = 6°$) for the Jurassic Chon Aike Formation from 54 virtual-geomagnetic-pole determinations from the Camerones Bay, Deseado and Estancia Reconquista districts. This reduces to a magnetic direction of 007.6, —66.2 at the latitude of Lago San Martin (Fig. 4A), and agrees well with our Lago San Martin result of 352, —57, and the determination of 002°, —62° for a Jurassic intrusion on Isla Rennel (Dalziel et al., 1973a).

Seven reconnaissance samples from the northern fiords yielded useable result (Fig. 3c) that average as a normal direction trending 316° at —64° (Table II). The samples come from three widely separated localities (Fig. 1); a tuff (Spec. 182) from a small islet in Seno Agostini, five samples of columnar-jointed ignimbrite (Specs. 299—303) from Puerto Toto in Almirantazgo, and one sample of foliated rhyolite (Spec. 295) from Cerro Senoret. The samples were corrected for different bedding dips — 76° for the massive tuff, 16° for the ignimbrite and 65° for the rhyolite — so that a fold test is achieved. The Fisher dispersion parameter increases from $K = 6$ to $K = 34$ so that the fold test is significant at the 99% level (McElhinny, 1964).

Cretaceous samples

The only Cretaceous rocks that yielded results come from Navarino Island where the Dientes de Navarino sill provided a major sampling site (Fig. 1). The petrology and geological setting of this sill have been described in detail by Katz and Watters (1966). The main sill is 200—300 m thick and composed of quartz dolerite with quartz, K-feldspar mesostasis; the country rock has undergone slight burial metamorphism (Watters, 1965) but nevertheless the sill is generally unaffected and expected to have retained the original magnetization. The country rocks are gently folded and broken by block faults (Fig. 2c). We consider it unlikely that such a thick and extensive series of dolerites could have been intruded after folding without penetrating the faults and some of the sills are themselves offset by faults (Katz and Watters, 1966). Consequently, tectonic dip corrections of 40° to 50° have been made

based on the dip of the contact and internal cooling-joint columns. The Yahgan formation is considered to be Cretaceous on the basis of *Inoceramus* (Katz and Watters, 1966). A pyroxene concentrate from the main sill has been dated as 115 ± 5 m.y. (Table I) which confirms that the sediments are Early Cretaceous or older.

Ten samples collected at intervals throughout the main sill and one sample (Spec. 124) collected from about 3 km to the west were measured. In addition, two samples (Specs. 125 and 134), from a thinner sill a few tens of metres below the main sill, from both these localities were measured (Figs. 1 and 2c). The results from all of these samples cluster together to give a mean direction trending 319° at −52° (Figs. 3 and 4). Samples of the greywacke country rock proved unreliable due to scatter as did those of the upper hornfels; a hornfelsed limestone (Spec. 119) near the lower contact of the sill, however, gave a magnetic direction of 335°, −35° (not included in the mean) which is close to the main cluster (Fig. 3e); this indicates that the dolerite has been magnetically stable since cooling.

Cainozoic samples

Dolerite sills and dykes, diorite and andesite sills, and basalt and rhyolite plugs were sampled at Lago San Martin. Basalt flows on Isla Carlos in the Straits of Magellan, dolerite dykes cutting the Jurassic in several places, and a major dyke and sill swarm on Sierra Valdivieso were also sampled (Figs. 1, 2a, b). Two to four specimens were collected from each rock body but many gave scattered or inconsistent results.

At Lago San Martin three specimens from each of the Cache Aike rhyolite plug (Specs. 1—3), Cerro Morro basalt plug (Specs. 7—9), and dolerite sills in the Cretaceous (Specs. 25—27) gave reasonably tight clustered directions (Fig. $3a_2$; Table II). A single specimen of a diorite sill (Spec. 5) in the Cretaceous rocks south of the lake gave a direction away from the other results (Fig. $3a_2$), and a very fresh andesite sill intruded at the junction between the Jurassic and Cretaceous sequence was unfortunately magnetically inconsistent.

Krank (1932) had noted that the Isla Carlos basalts were younger than the rocks on either side of the Straits of Magellan and equated them with the "laccolithic" intrusions of southern Patagonia. The basalts dip 20° northeastwards where sampled and the four directions obtained cluster away from the Lago San Martin results (Fig. 3b).

It was apparent that these rocks were not all the same age and three samples were dated. The Isla Carlos basalt (Spec. 47) yielded a whole rock age of 21.3 ± 0.6 m.y., the diorite sill (Spec. 5), a biotite age of 14.0 ± 0.5 m.y., and the andesite sill (Spec. 16), a hornblende age of 14.1 ± 0.5 m.y., (Table I). The plugs of Chache Aike and Cerro Morro are clearly very young. The spread of these Cainozoic results (Fig. $3a_2$, b) may be due, at least in part, to apparent polar wandering from the early Miocene towards the present day,

or to inadequate sampling. It is unlikely that the Isla Carlos basalts have been rotated, because Dalziel et al. (1973a) record a 334°-trending direction from a basic dyke northeast of Isla Rennel in the northward-trending belt (Fig. 4, site *f*).

Dolerite dykes from Peninsula Brunswick and the northern fiords of Tierra del Fuego gave inconsistent results, and the Cerro Verde plug (Spec. 314—316) and samples of dykes and a sill from Sierra Valdivieso gave scattered results (Figs. 1 and 3d); the best that can be said is that they fall into the same general field as the Lago San Martin directions (cf., Fig. 3a, 3d), and the meaned directions are similar (Table II; Fig. 4c).

Important samples of Palaeozoic(?) Cerro Svea Granite, folded Cretaceous sandstones from Isla Alta in Bahia Ainsworth, and a Cainozoic diorite dyke cutting the Dientes de Navarino sill unfortunately yielded inconsistent results.

DISCUSSION

The results of our palaeomagnetic determinations are summarized in Table II, and the azimuths of the mean directions for each locality are compared with the results of Dalziel et al. (1973a) on Fig. 4.

When the Jurassic direction for the northern Fiords (Fig. 4A, site *c*) is compared with the averaged directions from Lago San Martin (Fig. 4A, site *a*) or the Chon Aike Formation (Fig. 4A, site *i*), a rotation of some 40° of the eastward-trending belt from the northward-trending belt is clearly indicated, and the virtual geomagnetic poles are significantly displaced (Fig. 4D). In this connection the tectonic bedding correction made on the Navarino samples is critical, because the uncorrected directions trend southwestwards (238, —63, $K = 54$) parallel to those from the Santa Rosa pluton (Fig. 4B, site h) from western Navarino Island (Dalziel et al., 1973a). We conclude tentatively that there may have been some bending (25°) between the Jurassic and Cretaceous (Fig. 5a).

The Cainozoic directions from Lago San Martin (Fig. 4C, site *a*), Isla Carlos (Fig. 4C, site *b*) and Sierra Valdivieso (Fig. 4C, site *d*) are essentially parallel to each other and to the directions obtained in the outer arc (Fig. 4C, sites *f* and *g*), suggesting that bending ceased before the emplacement of these basic intrusives.

The structural bend of the orogen is 90°, or more where local complications give northeasterly strikes, yet our results from the inner arc indicate that no more than 48° of bending has occurred; this suggests that part of the curvature may be original (Fig. 5a). Dalziel et al. (1973a) obtained different results from the outer arc, where Middle of Late Cretaceous Andean granodiorites yielded magnetic directions trending northwards in the northward trending belt (Fig. 4B, site *f*) and southwestwards in the eastward trending belt (Fig. 4B, sites *g* and *h*). These results indicate that a bending of greater than 90° has occurred. It is perhaps significant that our sampling sites in the

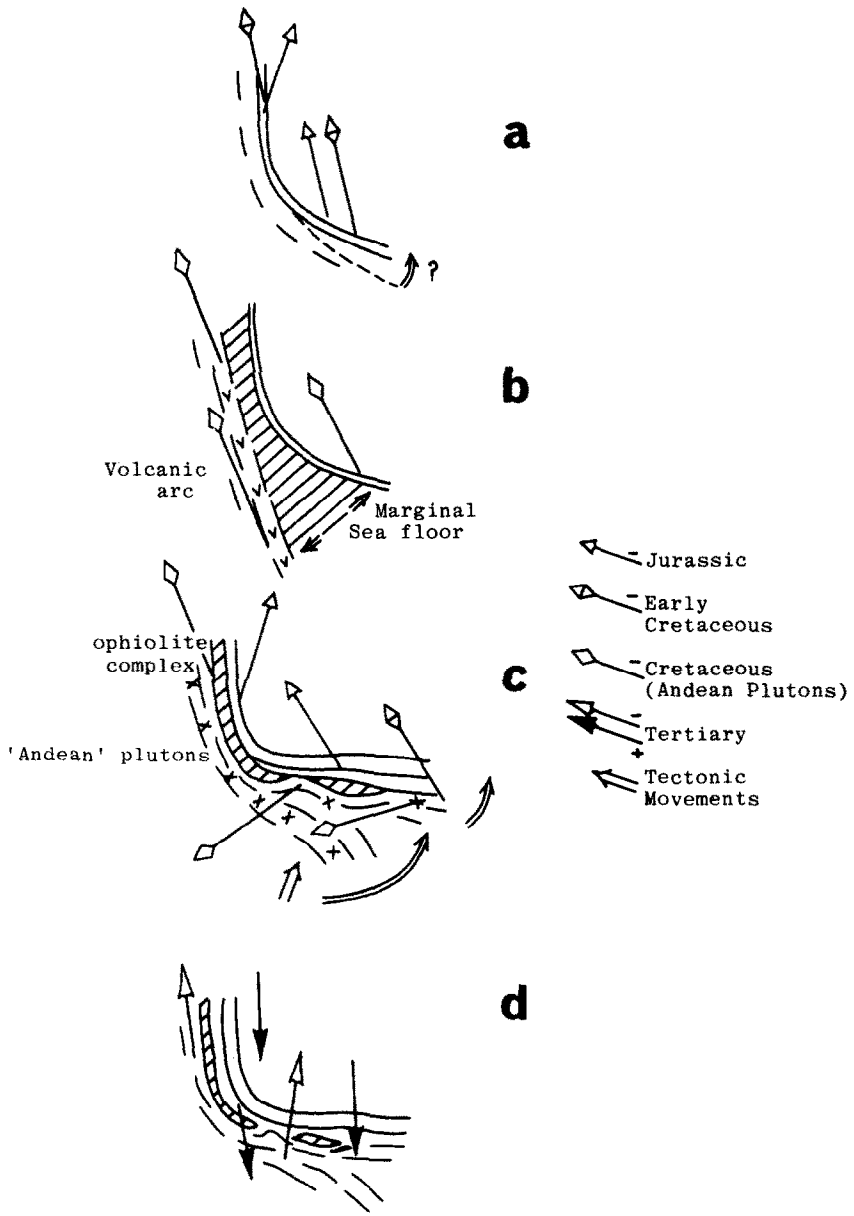


Fig. 5. Possible development of the Magellanes Orocline — displacement of magnetic directions by oroclinal bending of an initial Mesozoic volcanic arc and marginal basin. a. Jurassic—Early Cretaceous situation — possibly slight bending. b. Mid-Cretaceous volcanic arc separated by marginal sea. c. Late Cretaceous deformation closing marginal sea — bending with folding, thrusting and pluton intrusion. d. Cainozoic to Recent situation — bending ceases, intrusion of basic dykes, sills etc. Broad arrows show movements.

inner arc (Fig. 4A, sites *a* and *c*) with the exception of Navarino Island (Fig. 4B, site *e*), are separated from those of Dalziel et al. (1973a) in the outer arc (Fig. 4A, B, sites *f*, *g* and *h*) by a zone of Cretaceous greenstones (ophiolites) that have been interpreted as a collapsed marginal ocean basin (Dalziel et al., 1973b). The Dientes Sill (Fig. 4B, site *e*), however, lies on strike with the Santa Rosa pluton (Fig. 4B, site *h*) in northwestern Navarino Island but they are probably separated by a major N—S fault (Katz and Watters, 1966) so that this may not invalidate the hypothesis.

Bending is likely to have accompanied the folding and thrusting associated with the main Andean orogeny, and it is likely that movements in the colliding outer arc would be more complex than those in the inner foreland arc; this may also explain the divergence in trends south of Beagle Canal. A diagrammatic development sequence is presented in Fig. 5.

Bending of an already curved arc probably started in the Late Jurassic and continued with increasing intensity until after the emplacement of post-fold Andean plutons about 80 m.y. ago (Halpern and Rex, 1972; Dalziel et al., 1973a). Katz (1973) disputes the significance of the pluton date and argues that folding continued until the end of the Cretaceous. The Cainozoic palaeomagnetic determinations indicate that bending had ceased by Miocene times (Fig. 5d). Despite the possibility of post-fold bending (Dalziel et al., 1973a) there is no evidence of a major cross-folding on the inner side of the arc. Our structural studies indicate that bending has been accommodated by near-vertical rotation within and between thrust slices. The deformational aspects and kinematic interpretation of this orocline will be described elsewhere (Burns and Rickard, in prep.).

ACKNOWLEDGMENTS

We are extremely grateful to the numerous organizations and people who kindly assisted us in this project. The expedition was financed by the Australian Research Grants Committee. H.R. Katz (N.Z. Geological Survey) assisted us in the planning stages of the work. Prof. R. Zardini and E. Espisua of the University of Buenos Aires provided transport and assisted us with mapping at Lago San Martin. Dr. A. Riccardi of the Argentine Geological Survey lent his maps and aerial photographs of Lago San Martin. Transport to Isla Carlos and Peninsula Brunswick was by the Chilean Navy; accommodation and local transport there was provided by Sr. G. Ferretti of the Societie Minera de Cutter Cove. The Chilean Geological Survey (I.I.G.) and Oil Company (E.Na.P.) provided maps, reports and airphotographs, and A. Tobar (I.I.G.) and L. Silva and I. Galay (E.Na.P.) gave valuable field assistance. E.Na.P. provided transport on M.V. Ivan for the work in the nothern fiords of Tierra del Fuego, and Dr. M.W. McElhinny kindly allowed the use of palaeomagnetic equipment at the Australian National University. We are indebted to Prof. R. Zardini and Dr. J. Cosentino of the University of Buenos Aires, Dr. J. Corvalan (I.I.G. Chile) Drs. O. Schneider and E. Gonzalez

(E.Na.P. Chile) and Admiral Walbaum of the Third Naval Zone, Punta Arenas Chile for their major contribution to the success of the expedition. We are grateful also to Dr. I.W.D. Dalziel for discussions on his work in the region.

We would also like to thank Prof. D.A. Brown and Dr. M.W. McElhinny of the Australian National University for critically reading the manuscript.

The authors are pleased to have been able to present this paper to the Carey Symposium at the University of Tasmania in February, 1977 by way of tribute to Professor Carey's contributions to tectonics and the inspiration he has given to a generation of Australian geologists.

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